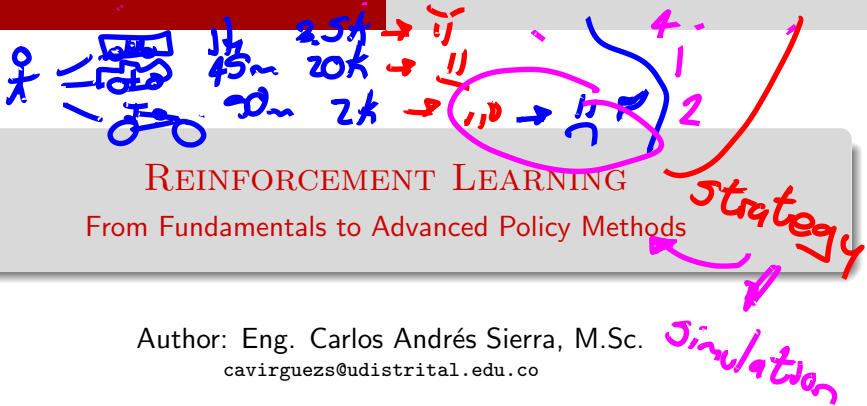




REINFORCEMENT LEARNING

From Fundamentals to Advanced Policy Methods

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supervised
unsupervised

2026-III



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FRANCISCO JOSÉ DE CALDAS

Outline

- 1 Introduction to Reinforcement Learning Fundamentals
- 2 Advanced Reinforcement Learning Algorithms
- 3 Policy Optimization Methods
- 4 Advanced Policy Methods & Multi-Agent RL

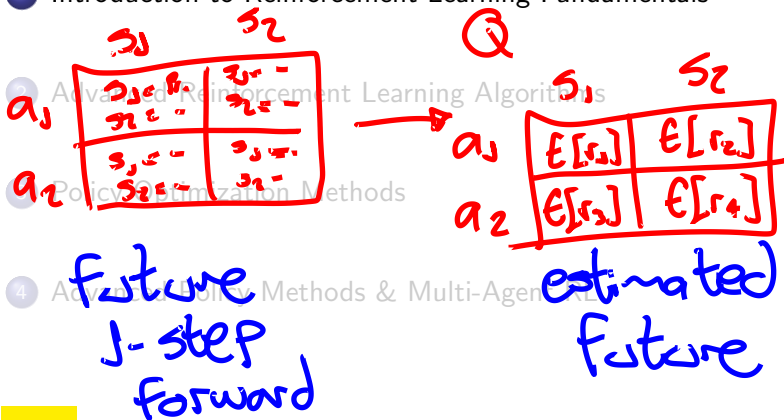
Alpha Go

Deep
Neural
Networks



Outline

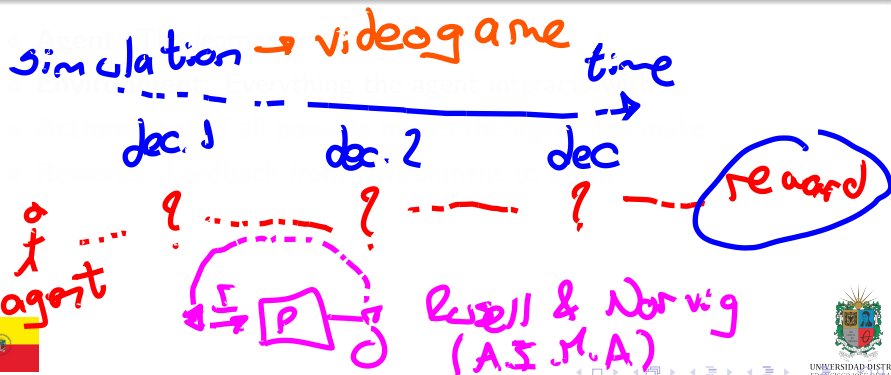
1 Introduction to Reinforcement Learning Fundamentals



What is Reinforcement Learning?

Reinforcement Learning

- **Reinforcement Learning** is a computational approach to learning whereby an **agent** tries to maximize the notion of cumulative **reward**.
- The agent learns to achieve a **goal** in an **uncertain**, potentially complex **environment**.



What is Reinforcement Learning?

Reinforcement Learning

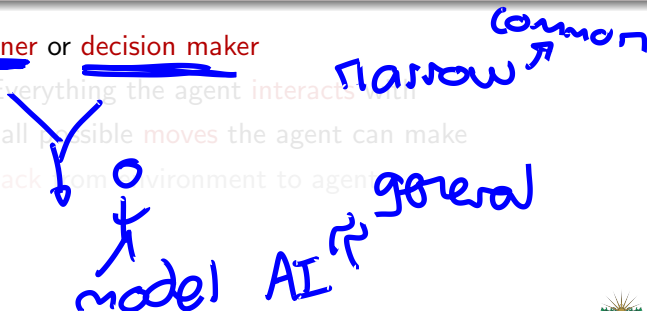
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- **Agent:** The **learner** or **decision maker**

- **Environment:** Everything the agent interacts with

- **Actions:** Set of all possible **moves** the agent can make

- **Rewards:** Feedback from environment to agent



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world
stages

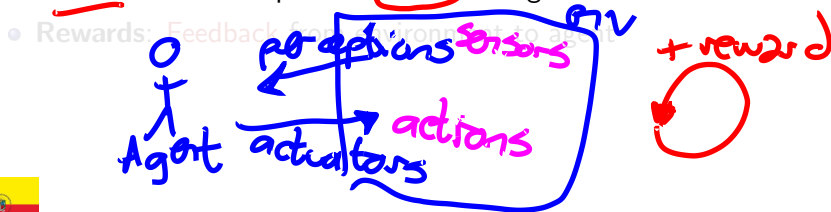
computational
simulation



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RL vs. Other Learning Paradigms

Supervised Learning:

- Uses **labeled** data
- **Direct feedback**
- **Static datasets**

Unsupervised Learning:

- No labels
- **Pattern discovery**
- Structure finding

data

Jan → ML 1.0

JUNE → retrain
data change → ML 2.0

J-dec → ML 3.0



RL vs. Other Learning Paradigms

Supervised Learning:

- Uses **labeled** data
- Direct feedback **X**
- **Static** datasets

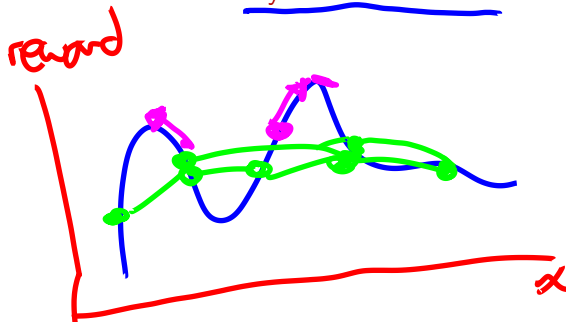
Unsupervised Learning:

- No labels
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Reinforcement Learning:

No data

- Trial-and-error learning
- Delayed rewards
- Sequential decision making
- Exploration vs. exploitation
- Dynamic environments



Markov Decision Processes (MDPs)

An MDP is defined by the tuple (S, A, P, R, γ) :

- **S**: Set of **states**
- **A**: Set of **actions**
- **P**: **Transition** probabilities $P(s'|s, a)$
- **R**: **Reward** function $R(s, a, s')$
- γ : **Discount** factor $[0, 1]$



jump
run
shoot

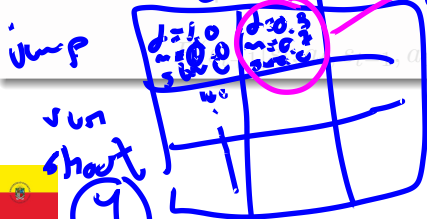
ACTIONS

die
~~lose~~ new
STATES
stay

→ max

TRANSITION

The future is dependent of the present state.



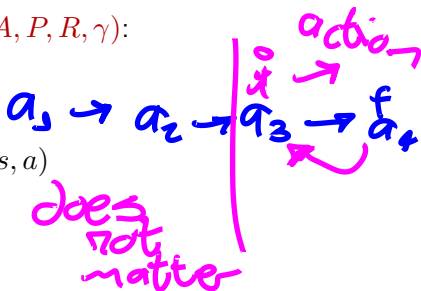
$f(x, y, p=s)$



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Markov Property

The future is independent of the past given the present state.

$$P(\underline{s_{t+1}} | \underline{s_t, a_t, s_{t-1}, a_{t-1}, \dots}) = P(\underline{s_{t+1}} | \underline{s_t, a_t})$$



Agent-Environment Interaction

The RL Loop:

- 1 Agent observes current **state** s_t
- 2 Agent selects **action** a_t based on policy $\pi(a|s)$
- 3 Environment transitions to new **state** s_{t+1}
- 4 Environment gives **reward** r_{t+1}
- 5 Agent updates its **knowledge**
- 6 Repeat...

$s_t = 0$ → start

learning
→ update probabilities

transitions
matrix
(s, a, s')
conditional
prob.

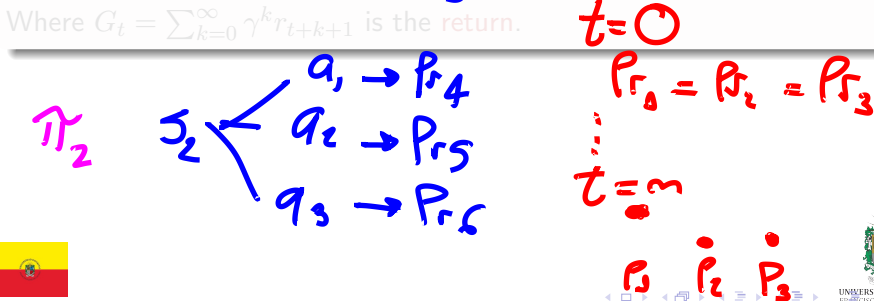
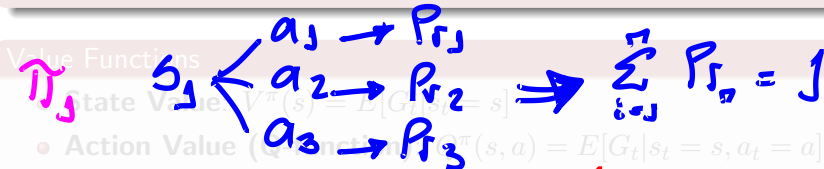
$$r_{\text{av}} = \sum_{t=0}^T r_t$$



Value Functions and Policies

Policy

A **policy** $\pi(a|s)$ is a mapping from states to probability distributions over actions.



Value Functions and Policies

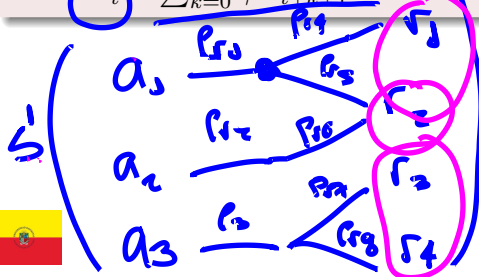
Policy

A **policy** $\pi(a|s)$ is a mapping from states to **probability distributions** over actions.

Value Functions

- **State Value:** $V^\pi(s) = E[G_t | s_t = s]$
- **Action Value (Q-function):** $Q^\pi(s, a) = E[G_t | s_t = s, a_t = a]$

Where $G_t = \sum_{k=0}^{\infty} \gamma^k r_{t+k+1}$ is the **return**.

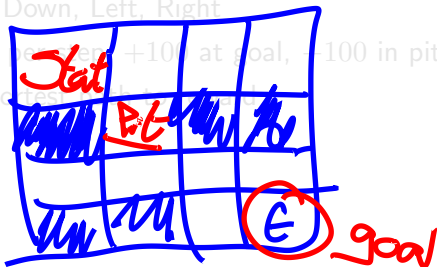


$avg(r_1, r_2, r_3, r_4)$
 $\downarrow$
 $E(G_t | s_t = s)$
 $E[G_t | s_t = s, a_t = a_1] = avg(r_1, r_2)$
 $E[G_t | s_t = s, a_t = a_2] = r_2$
 $E[G_t | s_t = s, a_t = a_3] = r_4$



Practical Example: Grid World Navigation

- **Environment:** 4×4 grid with walls and goal
- **States:** Agent's position (x, y)
- **Actions:** Up, Down, Left, Right
- **Rewards:** -1 per step, $+100$ at goal, -100 in pit
- **Goal:** Find shortest path



Practical Example: Grid World Navigation

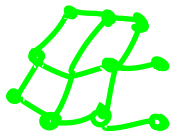
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Handwritten annotations: "Punish" with an arrow pointing to the -100 in the rewards, and "die" next to it. A green underline is under "Goal: Find shortest path to reward". A pink arrow points from "Rewards" to the word "design".



Q-Learning Algorithm I

Definition

Q-Learning is a **model-free** reinforcement learning algorithm that learns the **optimal action-value function** $Q^*(s, a)$ directly from interactions with the environment.

↓ policy
optimization

$$\max E[\pi(a|s)]$$



Q-Learning Algorithm II

Q-Learning Update Rule

$$Q(s_t, a_t) \leftarrow Q(s_t, a_t) + \alpha [r_{t+1} + \gamma \max_a Q(\overbrace{s_{t+1}}^{s'}) - Q(s_t, a_t)]$$

• α : Learning rate $[0, 1]$

• γ : Discount factor $[0, 1]$

• Off-policy: Learns optimal policy regardless of behavior policy

• Model-free: No need to know transition probabilities

$f(s, a, s') Q$

$s \begin{cases} a_1 - r_1 \\ a_2 - r_2 \\ \vdots \\ a_n - r_n \end{cases}$

$\max_{r_1} ? \rightarrow \text{action}$

max reward



Q-Learning Algorithm II *→ stable baselines (OpenAI)*

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actions
states | **AlphaGo**

random
or Uniform ✓



Bellman Equation for Q-Values

Bellman Optimality Equation

$$Q^*(s, a) = \underline{E}[r + \gamma \max_{a'} Q^*(s', a') | s, a]$$

s' → 1-step

2-steps forward

- Q-Learning approximates this equation through temporal difference learning
- The TD error: $\delta = r + \gamma \max_{a'} Q(s', a) - Q(s, a)$



Bellman Equation for Q-Values

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classic rule



Bellman Equation for Q-Values

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Exploration vs. Exploitation

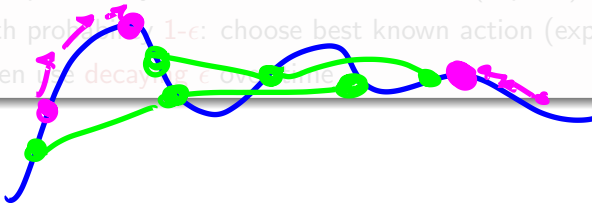
The Exploration-Exploitation Dilemma

Should the agent choose the best known action (exploit) or try something new (explore)?

jump off
local optimal

ϵ -Greedy Strategy

- With probability ϵ : choose random action (explore)
- With probability $1-\epsilon$: choose best known action (exploit)
- Often the decaying ϵ over time



Exploration vs. Exploitation

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- With probability $1-\epsilon$: choose best known action (exploit)
- Often use decaying ϵ over time

$\epsilon = 0.8 = \text{explore}$

$\text{exploit} = 0.2$

0.7
0.4
0.1

ca 3

0.6
0.9



Practical Example: Q-Learning in Grid World

- Initialize $Q(s, a)$ arbitrarily (e.g., all zeros)
- For each episode:
 - Initialize state s
 - Repeat until episode ends:
 - Choose action a using ϵ -greedy policy
 - Take action a , observe reward r and new state s'
 - Update Q-value: $Q(s, a) \leftarrow Q(s, a) + \alpha[r + \gamma \max_a Q(s', a) - Q(s, a)]$
 - Update state: $s \leftarrow s'$

np. zeros



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for $v = v.copy()$



Outline

How to learn

1 Introduction to Reinforcement Learning Fundamentals

2 Advanced Reinforcement Learning Algorithms

3 Policy Optimization Methods

4 Advanced Policy Methods & Multi-Agent RL

• no data
• experience
• policy, epsilon-greedy, ...



SARSA Algorithm

SARSA Update Rule

$$Q(s_t, a_t) \leftarrow Q(s_t, a_t) + \alpha[r_{t+1} + \gamma Q(s_{t+1}, a_{t+1}) - Q(s_t, a_t)]$$

- **On-policy**: Updates based on the actual policy being followed
- **SARSA**: State-Action-Reward-State-Action sequence
 - More conservative than Q-Learning
 - Better for safety-critical applications



SARSA Algorithm

SARSA Update Rule

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- **SARSA**: State-Action-Reward-State-Action sequence
- More **conservative** than Q-Learning $\rightarrow \sim \text{future}$
- Better for **safety-critical** applications

$\hookrightarrow$ does not mean
"the Fast"



On-Policy vs. Off-Policy Learning

On-Policy (SARSA):

- Learns about policy currently following
- More conservative
- Accounts for exploration in updates
- Safer in risky environments

Off-Policy (Q-Learning):

- Learns about optimal policy
- More aggressive
- Ignores exploration in updates
- Faster convergence

exploration ↑



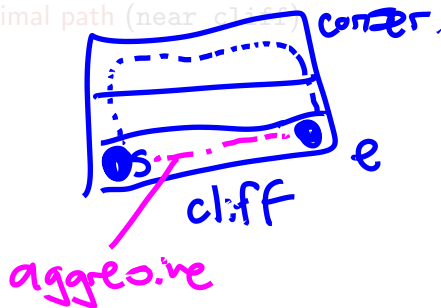
SARSA Algorithm Mechanics

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 - Choose next action a' using ϵ -greedy policy
 - Update Q-value: $Q(s, a) \leftarrow Q(s, a) + \alpha [r + \gamma Q(s', a') - Q(s, a)]$
 - Update state and action: $s \leftarrow s', a \leftarrow a'$



Practical Example: Cliff Walking Problem

- **Environment:** Agent walks along a cliff edge
- **Cliff:** Large **negative reward** (-100)
- **Goal:** Reach the end safely
- **SARSA:** Takes **safer path** (away from cliff)
- **Q-Learning:** Takes **riskier optimal path** (near cliff)



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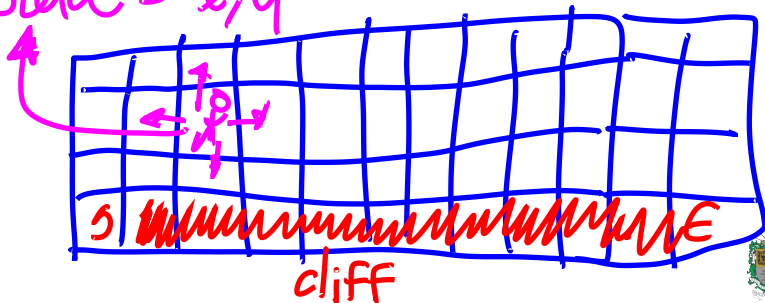


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$\epsilon = 0.1$
 0-0.1 / 0.1-0.1
 Exploit / Exploit

state = x, y



Function Approximation in RL

The Problem

- Large or continuous state spaces
- Cannot store Q-values for every state-action pair
- Need to generalize from seen to unseen states

Solution: Function Approximation

Use neural networks to approximate: $Q(s, a) \approx Q(s, a; \theta)$



Function Approximation in RL

 f_θ

The Problem

- Large or **continuous state spaces**
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 $Q(s, a; \theta)$

Solution: Function Approximation

Use **neural networks** to approximate: $Q(s, a) \approx Q(s, a; \theta)$

→ supervised learning

prediction



Deep Q Network (DQN) Architecture

↳ learn from environment ↳ most popular

Key Components

- Deep Neural Network: Approximates Q-function
- Experience Replay: Stores and samples past experiences
- Target Network: Stable Q-targets for training
- Fixed Q-Targets: Reduces correlation in updates

new neural networks

Loss Function

$$L(\theta) = E[(r + \gamma \max_{a'} Q(s', a'; \theta^-) - Q(s, a; \theta))^2]$$

memory
make decisions



Deep Q Network (DQN) Architecture

Key Components

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Handwritten annotations for the Loss Function:

- $\arg\max_{a'}$ (written in pink)
- $\max_{a'}$ (written in blue)
- $Q[s]$ (written in pink, with "next state" written above it)
- $Q(s')$ (written in blue)



Experience Replay Mechanism

- **Store:** Experiences (s, a, r, s') in replay **buffer**
- **Sample:** Random mini-batch for training
- **Benefits:**
 - Breaks temporal correlation
 - Improves data efficiency
 - Enables batch learning

partial, reward

current

future



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↓
subset < actions
states



DQN for Atari Breakout: Practical Example

Atari Breakout Implementation

- **Input:** Raw pixels ($210 \times 160 \times 3$) $\rightarrow$ preprocessed ($84 \times 84 \times 4$)
- **Network:** CNN with 3 conv layers + 2 dense layers
- **Actions:** 4 discrete actions (NOOP, FIRE, RIGHT, LEFT)

Preprocessing: Grayscale, resize, frame stacking (4 frames)
 Reward: +10 for each brick destroyed, +5 for clearing level

images
video



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- **Strategy:** Agent learns to tunnel through bricks for maximum score
- **Performance:** Achieves superhuman performance (400 points)

Handwritten notes: $0-255$ (twice), $\text{Colors} = R, G, B$, $0-255$ (twice)



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4 Multi-Agent Policy Methods & Multi-Agent RL

LA - HUST / ... / Austin - new journey / ... / NY - Miami / ...

variables $\leftarrow$ vol speed

graph $\Rightarrow$ Forecasting (1 day / 1-7 future)

NY & DC

3 days

max 11 hours

600 miles

2-3 weeks

Chain-of-loads strategy

RL DON Sunday



Policy-Based vs. Value-Based Methods

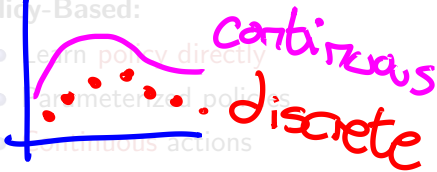
Value-Based:

- Learn **value functions** Q
- Derive policy from values
- Discrete action spaces
- Q-Learning, SARSA



Policy-Based:

- Learn policy directly
- Parameterized policies
- Continuous actions
- Stochastic policies



Policy-Based vs. Value-Based Methods

Value-Based:

- Learn **value functions**
- Derive policy from values
- Discrete action spaces
- Q-Learning, SARSA

no reward

Policy-Based:

- Learn **policy directly**
- Parameterized policies
- **Continuous** actions
- Stochastic policies

random
+
probabilities

robot
arm



Policy Gradient Theorem

Gradient of Expected Return

$$\nabla_{\theta} J(\theta) = E_{\pi_{\theta}} [\nabla_{\theta} \log \pi_{\theta}(a|s) Q^{\pi_{\theta}}(s, a)]$$

$\nabla \rightarrow \bigcirc$

error } $\rightarrow \bigcirc$
loss }

convergence



Policy Gradient Theorem

Gradient of Expected Return

$$\nabla_{\theta} J(\theta) = E_{\pi_{\theta}}[\nabla_{\theta} \log \pi_{\theta}(a|s) Q^{\pi_{\theta}}(s, a)]$$

- $\hat{\theta}$: Policy parameters
- **Intuition:** Move parameters in direction that **increases probability** of good actions
- **Key insight:** We can estimate gradients through **sampling**



REINFORCE Algorithm

REINFORCE Update

$$\theta \leftarrow \theta + \alpha \nabla_{\theta} \log \pi_{\theta}(a_t | s_t) G_t$$

- Uses **Monte Carlo** return G_t
- High **variance** but unbiased
- Often use **baseline** to reduce variance
- $\theta \leftarrow \theta + \alpha \nabla_{\theta} \log \pi_{\theta}(a_t | s_t) (G_t - b(s_t))$



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Actor-Critic Architecture

Two Components

- **Actor:** Policy $\pi_{\theta}(a|s)$ — **chooses** actions
- **Critic:** Value function $V_{\phi}(s)$ — **evaluates** actions

Benefits

- Lower variance than REINFORCE
- Online learning (no need to wait for episode end)
- Uses TD error: $\delta_t = r_{t+1} + \gamma V(s_{t+1}) - V(s_t)$



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Proximal Policy Optimization (PPO) Motivation

Problems with Basic Policy Gradients

- Large policy updates can be **destructive**
- Hard to choose appropriate **step size**
- Poor **sample efficiency**

PPO Solution

Constrain policy updates to stay in a **trust region** around current policy



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PPO Clipped Objective

Clipped Surrogate Objective

$$L^{CLIP}(\theta) = E[\min(r_t(\theta)\hat{A}_t, \text{clip}(r_t(\theta), 1 - \epsilon, 1 + \epsilon)\hat{A}_t)]$$

- $r_t(\theta) = \frac{\pi_\theta(a_t|s_t)}{\pi_{\theta_{old}}(a_t|s_t)}$ (probability ratio)
- $\hat{A}_t$ is advantage estimate
- Clips ratio to $[1 - \epsilon, 1 + \epsilon]$
- Prevents **destructively large** policy updates



Practical Example: PPO on LunarLander-v2

LunarLander-v2 Environment

- **State:** Position, velocity, angle, angular velocity, leg contact (8D continuous)
- **Actions:** 4 discrete actions (NOOP, LEFT, MAIN, RIGHT)
- **Goal:** Land safely between flags (+100 to +140 points)

• Reward Shaping: Distance, speed, angle penalties, fuel costs

• PPO Advantages: Stable learning with sparse rewards

• Policy Network: MLP ($8 \rightarrow 128 \rightarrow 128 \rightarrow 4$)

• Value Network: MLP ($8 \rightarrow 128 \rightarrow 128 \rightarrow 1$)

• Actor-Critic Architecture: Policy and Value networks share input

• Training: Iterative optimization of policy and value networks

• Evaluation: Performance measured by average score over multiple episodes

• Results: PPO achieves high performance with stable training



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Outline

- 1 Introduction to Reinforcement Learning Fundamentals
- 2 Advanced Reinforcement Learning Algorithms
- 3 Policy Optimization Methods
- 4 Advanced Policy Methods & Multi-Agent RL**



Trust Region Concept

Trust Region Optimization

- Optimize within a **trusted region** of current policy
- Ensures **monotonic improvement**
- Prevents policy **collapse**

Trust Region Policy Optimization (TRPO) Constraint

$$E[KL(\pi_{\theta_{old}}(\cdot|s), \pi_{\theta}(\cdot|s))] \leq \delta$$



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TRPO Mathematical Formulation

Constrained Optimization Problem

$$\text{maximize} \quad L(\theta) \quad (1)$$

$$\text{subject to} \quad \bar{D}_{KL}(\theta_{old}, \theta) \leq \delta \quad (2)$$

- Solved using conjugate gradient method
- More principled than PPO but computationally expensive
- Guarantees monotonic improvement



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Natural Policy Gradients

Natural Policy Gradient

- Uses **Fisher information matrix** to scale gradients
 - Accounts for **geometry** of policy space
 - Update: $\theta \leftarrow \theta + \alpha F^{-1} \nabla_{\theta} J(\theta)$
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- More **stable** updates than vanilla policy gradients
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Practical Example: TRPO in Complex Continuous Control

Complex Continuous Control Tasks

- **High-dimensional** state and action spaces
- Requires **precise** control and coordination
- Examples: *robotic manipulation, locomotion*

- TRPO Advantages: Handles complex dynamics and constraints
- Policy Network: Deep neural networks for representation
- Training: Sample-efficient with trust region updates
- Performance: State-of-the-art results



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Multi-Agent Environments

Challenges

- **Non-stationary** environments
- Other agents are **learning** simultaneously
- **Coordination** and competition
- **Communication** between agents

• **Cooperative:** Agents share common goal

• **Competitive:** Zero-sum games

• **Cooperative and competitive environments**



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Nash Equilibrium in RL

Nash Equilibrium

A strategy profile where no agent can improve by **unilaterally changing** their strategy.

- Difficult to compute in practice
- May not always exist or be unique
- Agents may not converge to equilibrium
- Alternative: Find correlated equilibria



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Centralized Training, Decentralized Execution

CTDE Paradigm

- **Training:** Use **global information** and centralized critic
- **Execution:** Each agent acts with only **local observations**

- Addresses partial observability
- Enables coordination during training
- Maintains decentralized execution
- Examples: MADDPG, COMA, QMIX



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Communication in Multi-Agent Systems

Communication Protocols

- **Explicit:** Agents exchange messages
- **Implicit:** Agents infer others' intentions from actions

• Communication can enhance coordination and performance

• Challenges: Bandwidth constraints, noise, security

• Communication can be used to coordinate actions and share information



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Thanks!

Questions?



Repo: <https://github.com/EngAndres/ud-public/tree/main/courses/reinforcement-learning>

